

Lab Exercise Steps & Solutions

Step 2: Change value of parameter `control_file_record_keep_time` to 21

1. System Level Alteration

```
show parameter control_file_record_keep_time
```

```
SQL> show parameter control_file_record_keep_time
```

NAME	TYPE	VALUE
control_file_record_keep_time	integer	7

```
SQL> alter session set control_file_record_keep_time=21;
```

```
alter system set control_file_record_keep_time=21;
```

```
SQL> alter system set control_file_record_keep_time=21 ;  
System altered.
```

2. Session Level Alteration Attempt

```
alter session set control_file_record_keep_time=21;
```

```
SQL> alter session set control_file_record_keep_time=21 ;  
alter session set control_file_record_keep_time=21  
      *  
ERROR at line 1:  
ORA-02096: specified initialization parameter is not modifiable with this  
option
```

Architectural Analysis: Why did the ORA-02096 error appear?

The `control_file_record_keep_time` parameter dictates how many days backup history is retained inside the physical control files before being overwritten. Because there is only **one shared set of control files** for the entire database instance, Oracle blocks individual users from altering this behavior locally. It is structurally constrained as a global setting and can only be altered using `ALTER SYSTEM`.

Step 3: Show value of SGA_TARGET, Try to change it to 8G. What did you notice?

1. View Current Status

```
show parameter sga_target
```

```
SQL> show parameter sga_target
```

NAME	TYPE	VALUE
sga_target	big integer	2G

2. Session Level Alteration Attempt

```
alter session set sga_target=8G;
```

```
SQL> alter session set sga_target=8G;
alter session set sga_target=8G
*
ERROR at line 1:
ORA-02096: specified initialization parameter is not modifiable with this
option
```

same previous error, sga_target is a system-level global setting too.

3. System Level Alteration

```
alter system set sga_target=8G;
```

```
SQL> alter system set sga_target=8G;
alter system set sga_target=8G
*
ERROR at line 1:
ORA-02097: parameter cannot be modified because specified value is invalid
ORA-00823: Specified value of sga_target greater than sga_max_size
```

Observations:

The system explicitly rejected the modification. This occurred because **8G** exceeds the current memory ceiling profile defined by **sga_max_size**.

Step 4: How to solve the error of step 3?

To resolve this conflict, the hard boundary limit (`sga_max_size`) must be expanded first. However, attempting a direct dynamic modification fails:

```
alter system set sga_max_size=8G;
```

```
SQL> alter system set sga_max_size=8G;
alter system set sga_max_size=8G
                *
ERROR at line 1:
ORA-02095: specified initialization parameter cannot be modified
```

The Solution Workflow:

`sga_max_size` is a **static parameter**, meaning its value sets the shared memory block size allocated by Linux at boot time. It cannot be altered in active memory. To change it, you must update the static Server Parameter File (`SPFILE`) and recycle the instance:

```
-- 1. Write the new boundary size to the persistent SPFILE profile
alter system set sga_max_size=4G scope=spfile;
```

```
SQL> alter system set sga_max_size=4G scope=spfile;

System altered.
```

```
-- 2. Cleanly restart the instance to request the new allocation from Linux
SQL> shutdown immediate;
Database closed.
Database dismounted.
Oracle instance shut down.

SQL> startup;
ORACLE instance started.
Database mounted.
Database opened.
```

after the startup , we can change the `sga_target` to 8G.